

4 Exploratory Data Analysis and Visualization (Mendley Dataset)

This section presents an exploratory data analysis (EDA) of a blood cell image dataset used for automated blood cell classification. The dataset consists of microscopic images organized into folders, where each folder corresponds to a specific type of blood cells. The main goal of this analysis is to understand the quality and structure of the data by examining the correctness of the labels, the distribution of classes, the sizes of the images and the characteristics of the pixel-level. These insights help guide pre-processing steps and model training.

4.1 Data Quality Assessment

The dataset is arranged so that each folder represents a different types of blood cells and that all images inside a folder share the same label. This structure ensures that the dataset is fully labeled and suitable for supervised learning.

A detailed data quality check was carried out, leading to the following observations:

- No missing or incorrect labels were found.
- All image files paths were valid and accessible.
- No corrupted image files were detected.
- Only one image could not be read during resolution analysis and was removed from the dataset.

Overall, the dataset is clean and well-organized.

4.2 Class Distribution

The dataset contains eight different classes of blood cells, including neutrophils, eosinophils, basophils, lymphocytes, monocytes, immature granulocytes, erythroblasts, and platelets. The analysis of class frequencies indicates a noticeable class imbalance. We observe that Neutrophils represent the most largest class, while lymphocytes and basophil constitute the smallest classes.

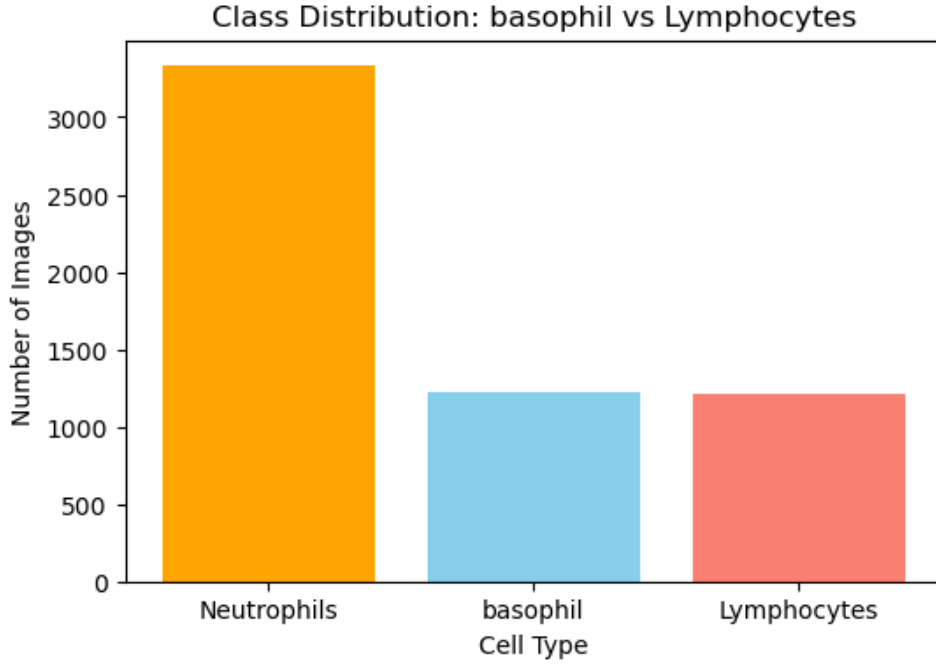
Statistically:

Mean number of samples per class: 2136.62

Standard deviation of class counts: 833.42

Imbalance ratio (max / min): 2.74

The mean number of samples per class provides an estimate of the average representation across blood cell types. The relatively high standard deviation of class counts indicates substantial variability in class sizes (a high variance among classes), confirming the presence of class imbalance. Furthermore, the imbalance ratio, defined as the ratio between the largest and smallest class, highlights a significant skew in the dataset distribution. Such imbalance may bias the learning process toward dominant classes, motivating the use of data augmentation during model training.



4.3 Image Resolution Distribution

The analysis of image resolutions shows that the dataset contains a variety of image sizes. Where the images have many different width–height combinations and most are around 360×360 , but not all.

This variety in images sizes means that some pre-processing is necessary. Therefore, it is recommended to resize all images to a standard resolution while keeping their aspect ratios, to maintain consistency and avoid any distortion. Moreover, considering the image resolution statistics:

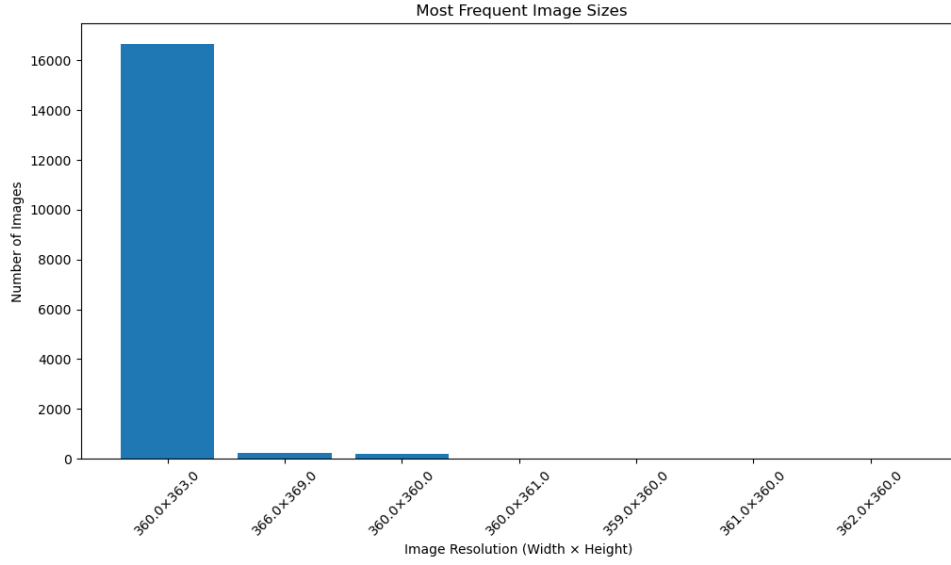
Mean width: 360.09 pixels

Mean height: 363.05 pixels

Standard deviation of width: 0.72 pixels

Standard deviation of height: 0.79 pixels

we confirm that the average image resolution was found to be 360.09×363.05 pixels. The standard deviations of width and height were 0.72 pixels and 0.79 pixels, respectively, indicating extremely low variability in image dimensions. This suggests that the images are nearly uniform in size. Consequently, resizing all images to a fixed resolution is statistically justified and is unlikely to introduce geometric distortion or loss of relevant cellular features.

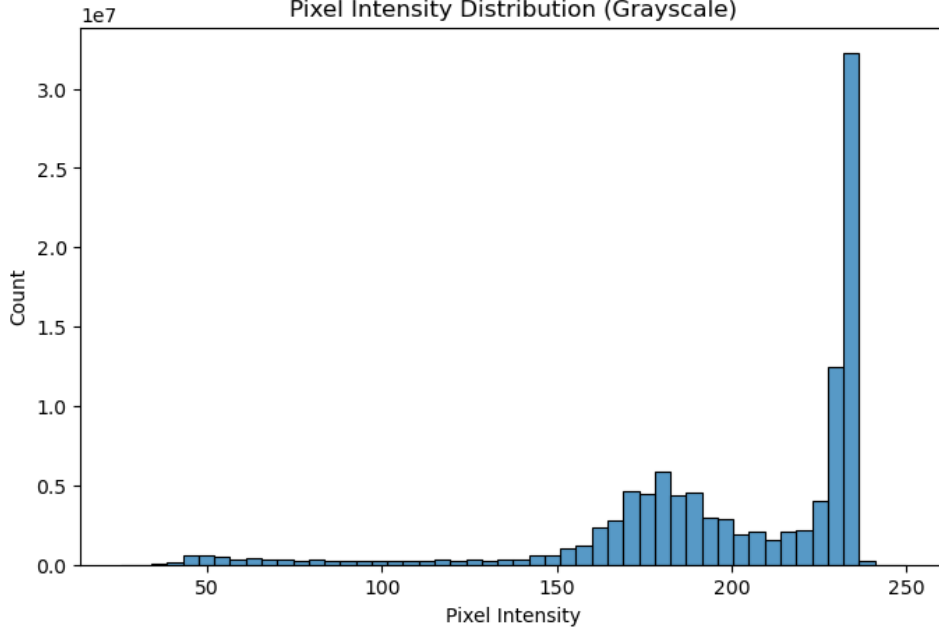


Width	Height	Count
360.0	363.0	16639
366.0	369.0	250
360.0	360.0	198
360.0	361.0	2
359.0	360.0	1
361.0	360.0	1
362.0	360.0	1

4.4 Pixel Intensity Distribution

To assess overall image contrast and illumination quality, pixel intensity values were examined by converting a representative sample of images to grayscale. The resulting intensity distribution covers the full range of values from 0 to 255.

The majority of pixel values are around mid-range intensities, with no extremely dark or bright pixels. This suggests that the images are well-exposed and possess sufficient contrast to distinguish important cellular structures such as nuclei, cytoplasm, and background.



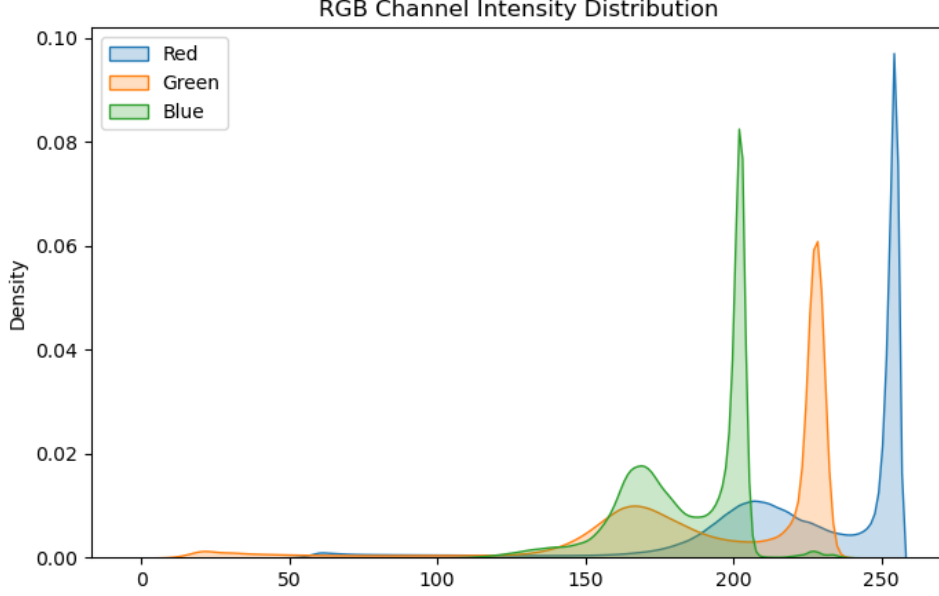
Regarding the mean intensity and Std intensity, we compute mean and std per image, then average them to get:
mean intensity (average brightness): 199.70 and
mean standard deviation (average contrast): 40.58

The average grayscale intensity across the dataset is 199.70, indicating generally bright images. The mean standard deviation of intensities is 40.58, reflecting sufficient contrast between cellular structures and the background. These findings suggest that the images are well illuminated and preserve meaningful intensity variations necessary for effective feature extraction in blood cell classification.

4.5 RGB Channel Analysis

An analysis of the individual RGB color channels was conducted to evaluate color information within the images. The red channel exhibits higher intensity values compared to the green and blue channels, while the blue channel shows greater density in lower to mid-intensity ranges. The green channel lies between the red and blue channels in terms of intensity distribution.

These observations are consistent with typical blood smear staining techniques. RGB preserves color variations, which help machine learning models distinguish between different cellular components.



RGB Channel Intensity Statistics:

Mean Red intensity: 222.83

Mean Green intensity: 190.91

Mean Blue intensity: 184.00

Std Red intensity: 38.97

Std Green intensity: 45.92

Std Blue intensity: 19.38

The RGB channel intensity analysis reveals distinct brightness and contrast characteristics across color channels. The red channel exhibits the highest mean intensity (222.83), followed by the green (190.91) and blue (184.00) channels, reflecting typical blood smear staining patterns. The green channel shows the highest standard deviation (45.92), indicating strong intensity variability and structural detail, while the blue channel demonstrates lower variability (19.38), corresponding to localized nuclear regions. These differences confirm that each RGB channel captures complementary information, justifying the use of color images for blood cell classification.

The exploratory data analysis demonstrates that the blood cells images dataset is clean, fully labeled, and visually informative. While the dataset is well-suited for machine learning applications, challenges such as class imbalance and image resolution variability must be addressed through appropriate pre-processing.

By applying image resizing and data augmentation techniques, the dataset can be effectively prepared for robust and accurate blood cell classification models.

4.6 Dataset Pre-processing

The dataset showed a clear class imbalance, with basophil and lymphocyte being the smallest classes, each comprising about 7% of the total samples. To reduce bias to-

ward majority classes and improve minority-class representation, data augmentation was applied only to these two classes.

Augmentation consisted of biologically plausible geometric transformations, including random rotations, horizontal and vertical flipping, and zooming. These operations preserve the structural characteristics of blood cells while increasing sample diversity. Each original image was augmented four times, increasing the number of basophil samples from 1,218 to 7,308 and lymphocyte samples from 1,214 to 7,284.

As a result, the representation of both classes increased to approximately 25% of the dataset, substantially reducing class imbalance. This targeted augmentation strategy is expected to improve model robustness and classification performance, particularly for previously underrepresented cell types.

Class	Count	Percentage
basophil	7308	24.98%
lymphocyte	7284	24.90%
neutrophil	3330	11.38%
eosinophil	3117	10.66%
ig	2895	9.90%
platelet	2348	8.03%
erythroblast	1551	5.30%
monocyt	1420	4.85 %

